

Module 1

Introduction: Scope, objectives and phases in materials management.

Procurement: purchase procedure, tender, earnest money, security deposit, purchase order, vendor rating. Receipt: Invoice, cash memo, inspection.

Storage: methods of storage.

Selective control techniques of inventory – ABC & VED analysis.

Inventory Theory: objectives of keeping inventory, structure of inventory problems and their analysis, relevant cost.

1. INTRODUCTION TO MATERIALS MANAGEMENT

- **Meaning of Materials Management**
- Materials Management is the **planning, purchasing, storing, moving, and controlling of materials** so that the **right material** is available in the **right quantity**, at the **right time**, at the **right place**, and at the **right cost**.
- ☞ Materials may include:
 - Raw materials
 - Spare parts
 - Components
 - Tools
 - Finished goods
- **Simple example:**
In a college canteen, materials management ensures that rice, vegetables, gas cylinders, and plates are available every day without excess waste or shortage.

Scope of Materials Management

- The scope covers the **entire life cycle of materials**:
- Forecasting material requirements
- Purchasing and procurement
- Receiving and inspection
- Storage and preservation
- Inventory control
- Disposal of scrap and obsolete items
- **Example:**
In a hospital, materials management covers medicines purchase, storage in proper conditions, issuing to wards, and disposal of expired drugs.

Objectives of Materials Management

- The main objectives are:
- **Uninterrupted production** – no shortage of materials
- **Minimum investment in inventory** – avoid blocking money
- **Reduction in wastage and losses**
- **Improved quality of materials**
- **Lower purchasing and storage costs**
- **Better supplier relationships**
- **Example:**
A factory storing too much steel wastes money on storage and rusting. Proper materials management avoids this.

Phases of Materials Management

- Materials management has **four main phases**:
- **Planning** – deciding what, how much, and when to buy
- **Procurement** – purchasing materials
- **Receiving & Inspection** – checking quantity and quality
- **Storage & Inventory Control** – safe keeping and issue

2. PROCUREMENT

- **Meaning of Procurement**
- Procurement is the process of **acquiring materials** at the **best price, best quality,** and **best delivery time** from reliable suppliers.
- **Purchase Procedure**
- The **purchase procedure** involves the following steps:
 - **Purchase requisition** – request from department
 - **Enquiry to suppliers**
 - **Receiving quotations**
 - **Comparison of quotations**
 - **Selection of supplier**
 - **Issue of purchase order**
 - **Follow-up**
 - **Receiving materials**
- **Example:**
Workshop requests 100 bearings → purchase department gets quotes → selects best vendor → issues purchase order.

- **Tender**
- A **tender** is a formal invitation to suppliers to submit quotations.
- **Types of Tenders:**
- **Open tender** – open to all suppliers
- **Limited tender** – sent to selected suppliers
- **Single tender** – only one supplier
- **Example:**
Government college buying lab equipment uses open tender.

- **Earnest Money Deposit (EMD)**
- A **small amount paid by bidder** along with tender
- Shows **seriousness of supplier**
- Returned to unsuccessful bidders
- **Example:**
₹10,000 deposited while bidding for a ₹5 lakh supply contract.
- **Security Deposit**
- Amount deposited by **selected supplier**
- Ensures **proper performance of contract**
- Refunded after successful completion
- **Example:**
5% of order value deposited till warranty period ends.

- **Purchase Order**
- A **purchase order (PO)** is a legal document issued to supplier containing:
 - Description of material
 - Quantity
 - Price
 - Delivery date
 - Payment terms
- **Example:**
PO for 500 kg copper wire at ₹750/kg.

- **Vendor Rating**
- Vendor rating evaluates supplier performance based on:
 - Quality
 - Delivery time
 - Price
 - Service
 - Reliability
- **Example:**

Vendor A delivers on time with good quality → high rating → preferred supplier.

3. RECEIPT OF MATERIALS

- **Invoice**
 - Document sent by supplier
 - Shows quantity, price, taxes, total amount
 - **Example:**
Invoice showing 100 motors × ₹2,000 each.
- **Cash Memo**
 - Proof of **immediate cash payment**
 - Used for small purchases
 - **Example:**
Buying office stationery locally.
- **Inspection**
 - Inspection ensures materials received are:
 - Correct quantity
 - Required quality
 - As per specifications
 - **Example:**
Checking diameter and strength of steel rods before acceptance.

- **4. STORAGE**
- **Meaning of Storage**
- Storage means **safe keeping of materials** until they are required.
- **Methods of Storage**
- **Centralized storage** – one main store
- **Decentralized storage** – department-wise stores
- **Open storage** – bulky items (cement, sand)
- **Closed storage** – costly or delicate items
- **FIFO storage** – first-in-first-out
- **Example:**
Medicines stored using FIFO to avoid expiry.

5. SELECTIVE INVENTORY CONTROL TECHNIQUES

- **ABC Analysis**
- ABC analysis classifies items based on **annual consumption value**.
- **Example:**
- A: Engines
- B: Bearings
- C: Nuts and bolts

Category	Value	Control
A	High value (10% items = 70% value)	Strict
B	Medium value	Moderate
C	Low value (70% items = 10% value)	Simple

- **1. What is ABC Analysis?**

- ABC Analysis is a selective inventory control technique used in Materials Management to classify items based on their annual consumption value.
- ☞ The idea is simple:
- Not all inventory items are equally important.
- Some items cost a lot and need strict control, while others cost very little and need only basic attention.
- ABC Analysis is based on the Pareto Principle (80–20 rule):
- 20% of items account for about 80% of total inventory value
- The remaining items contribute much less value

- **2. Objective of ABC Analysis**

- The main objectives are:
- To focus managerial attention on important items
- To reduce inventory cost
- To avoid overstocking and understocking
- To improve inventory control efficiency

- **3. Basis of ABC Analysis**

- Items are classified based on:
- $\text{Annual Consumption Value} = \text{Annual Quantity Used} \times \text{Unit Price}$

- **4. Classification of Items in ABC Analysis**
- Inventory items are divided into three categories:
- **A Items (Most Important)**
 - About 10–20% of total items
 - Account for 70–80% of total inventory value
 - High-value items
 - Control required:
 - Strict control
 - Frequent review
 - Accurate records
 - Top management attention

- **B Items (Moderately Important)**
- About 20–30% of items
- Account for 15–25% of inventory value
- Control required:

- Moderate control
- Periodic review
- Middle management supervision

- **C Items (Least Important)**
- About 50–70% of items
- Account for only 5–10% of inventory value
- Low-cost items
- Control required:

- Simple control
- Bulk purchasing
- Minimum record keeping

- **5. Steps Involved in ABC Analysis**

- **Step 1:** List all inventory items

- Prepare a list of all materials used in the organization.

- **Step 2:** Calculate annual consumption value

- Annual Consumption Value = Annual Usage × Unit Cost

- **Step 3:** Arrange items in descending order

- Sort items from highest value to lowest value.

- **Step 4:** Calculate cumulative value and percentage
- Find total inventory value
- Calculate cumulative percentage of value
- **Step 5:** Classify items into A, B, and C
- Based on cumulative percentages:
- First ~70–80% → A items
- Next ~15–25% → B items
- Remaining ~5–10% → C items

6. Numerical Example

Item	Annual Usage	Unit Cost (₹)	Annual Value (₹)
X	100	500	50,000
Y	200	150	30,000
Z	500	20	10,000
P	1000	5	5,000
Q	2000	2	4,000

Step 1: Arrange in descending order of value

Item	Annual Value (₹)	Cumulative %	Category
X	50,000	50%	A
Y	30,000	80%	A
Z	10,000	90%	B
P	5,000	95%	C
Q	4,000	100%	C

- **7. Advantages of ABC Analysis**

- Better inventory control
- Reduction in working capital
- Helps in efficient purchasing
- Improves management decision-making
- Saves time and cost

- **8. Limitations of ABC Analysis**

- Considers only cost, not criticality
- Not suitable where items are equally important
- Needs accurate data
- Does not consider lead time or scarcity
-  This is why ABC is often combined with VED analysis.

VED Analysis

- Based on **criticality** of items:
- **V – Vital** (no substitute)
- **E – Essential** (temporary shortage acceptable)
- **D – Desirable** (least important)
- **Example (Hospital):**
- V: Oxygen cylinders
- E: Gloves
- D: Furniture

- VED Analysis is a selective inventory control technique used to classify materials based on their criticality to production or service.
- VED stands for:
 - V – Vital
 - E – Essential
 - D – Desirable
- ➡ Unlike ABC analysis (which is based on cost/value), VED is based on importance and necessity.

- **2. Objective of VED Analysis**

- The main objectives are:
- To ensure continuous production/service
- To avoid shutdowns due to non-availability of critical items
- To prioritize items based on importance, not cost
- To improve inventory planning in critical systems

- **3. Basis of Classification in VED Analysis**

- Items are classified based on:
- Impact on production if the item is not available
- Availability of substitutes
- Lead time for procurement
- Effect on safety, quality, or operation

- **4. Categories in VED Analysis**

- **1. V – Vital Items**

- Absence stops production completely
- No substitute available
- Very critical items
- **Control required:**
- Zero stock-out allowed
- Continuous monitoring
- High safety stock
- Top management attention

- **Examples:**
- Main motor of a machine
- Critical bearing in a production line
- Life-saving drugs in a hospital

- **2. E – Essential Items**

- Absence partially affects production
- Substitutes may be available
- Delay can be tolerated for a short time
- Control required:
 - Moderate safety stock
 - Periodic review
 - Middle-level management control
- Examples:
 - Spare belts
 - Standard tools
 - Secondary components

- **3. D – Desirable Items**

- Absence does not affect production immediately
- Items improve convenience or efficiency
- Can be postponed or avoided

- Control required:
- Low safety stock
- Simple control
- Bulk purchasing if needed

- Examples:

- Decorative fittings
- Office stationery
- Optional accessories

- **5. Steps in VED Analysis**

- Step 1: Prepare list of items
- List all materials and spares used.
- Step 2: Assess criticality
- Evaluate each item based on:
 - Effect of non-availability
 - Lead time
 - Availability of substitutes
- Step 3: Classify items
- Group items into:
 - Vital
 - Essential
 - Desirable
- Step 4: Apply control measures
- Decide stock levels and monitoring intensity.

Example: Machine Spare Parts

Item	Effect if not available	Category
Main motor	Machine stops	Vital
Drive belt	Production slows	Essential
Indicator lamp	No major effect	Desirable

Difference Between ABC and VED (Quick Comparison)

Aspect	ABC Analysis	VED Analysis
Basis	Cost / Value	Criticality
Focus	Cost control	Continuity
Categories	A, B, C	V, E, D
Risk handled	Financial	Operational

6. INVENTORY THEORY

- **Objectives of Keeping Inventory**

- Smooth production
- Avoid stock-outs
- Meet sudden demand
- Economical purchasing
- Buffer against delays

- **Example:**

Maintaining spare parts for machines to avoid breakdown delays.

- **Structure of Inventory Problems**

- Inventory problems involve:

- **Demand rate**
- **Order quantity**
- **Reorder level**
- **Lead time**
- **Costs involved**

- **Example:**

How many units to order and when to order?

- **Relevant Costs in Inventory**
- **1. Ordering Cost**
- Cost of placing an order
- Includes paperwork, communication, inspection
- **Example:**
₹500 per order placed.
- **2. Carrying (Holding) Cost**
- Cost of storing inventory
- Includes rent, insurance, deterioration
- **Example:**
Warehouse rent, electricity, security.

- **3. Purchase Cost**

- Cost of buying materials

- **Example:**

₹100 per unit × 1,000 units.

- **4. Shortage Cost**

- Cost due to stock-out
- Loss of sales, production stoppage

- **Example:**

Machine idle due to unavailability of spare part.

- Materials management helps organizations **reduce cost, improve efficiency, and ensure smooth operations.** Proper purchasing, storage, inventory control, and cost analysis are essential for successful operations in industries, hospitals, and institutions.